Calendar — Question Pattern Catalog

Every Way Placement Exams Phrase a Calendar Question, Grouped by the Trick That Solves It

Aptitude Grind

Calendar questions look varied, but almost every one of them reduces to a handful of tricks from `02-fast-tricks-and-shortcuts.tex`. This file catalogues the actual **phrasings** you'll meet in TCS NQT, Infosys, Wipro, Capgemini, and CAT-style tests, grouped by category, each with the fastest method and a verified mini-example. Skim this before a test to pattern-match a question to its solution method in seconds.

1 Category A — Direct Day Lookup

A1. Find the Day of a Given Date

"What day of the week was 15 August 1947?"
The most common calendar question by far — direct application of the odd-days / anchor method. **Method:** §1 of Notes 02, or the mod-400 method in Notes 01.
Answer: 15 August 1947 was a **Friday**.

2 Category B — Relative Day Arithmetic (No Calendar Needed)

B1. Day After N Days

"Today is Tuesday. What day will it be after 100 days?"
Method: N mod 7, step forward. $100 \bmod 7 = 2 \rightarrow$ Tuesday $+2 =$ **Thursday**.

B2. Day Before N Days

"100 days before Friday is which day?"
Method: N mod 7, step backward. $100 \bmod 7 = 2 \rightarrow$ Friday $-2 =$ **Wednesday**.

B3. Yesterday / Tomorrow Questions

"Tomorrow is Monday. What was yesterday?"
Method: Anchor on the one day you're told, count off by ± 1. Tomorrow = Monday $\Rightarrow$ Today = Sunday $\Rightarrow$ Yesterday = **Saturday**.

B4. Chained Relative-Day Questions

"Three days after tomorrow is Friday. What day is today?"
Method: Work backward. "Three days after tomorrow" = today $+4$ = Friday $\Rightarrow$ today = Friday -4 = **Monday**.

B5. Mirror-Day Questions

"If yesterday was Thursday, what day will it be 100 days after tomorrow?"
Method: Pin down today first (yesterday = Thursday $\Rightarrow$ today = Friday), then tomorrow = Saturday, then apply B1: $100 \bmod 7 = 2 \rightarrow$ Saturday $+2$ = **Monday**.

B6. Calendar + Clock Combined

"It is Monday, 11 PM. What day and time will it be after 30 hours?"
Method: Split into full days + leftover hours: $30 = 24 + 6$. One full day moves Monday $\rightarrow$ Tuesday; the remaining 6 hours push 11 PM past midnight, landing on **Wednesday, 5 AM**. Always check whether leftover hours cross midnight.

3 Category C — Nth / Last Weekday of a Month

C1. Nth Weekday of a Month

"The third Monday of July 2026 falls on which date?"
Method: Find the weekday of the 1st, add 7 per subsequent occurrence. 1 July 2026 = Wednesday $\rightarrow$ first Monday = 6th $\rightarrow$ third Monday = $6 + 14$ = **20th**.

C2. Last Weekday of a Month

"Find the last Friday of February 2026 (28-day, non-leap)."
Method: Find the weekday of the last day, count back in steps of 7 to the target weekday.
Answer: 27 February 2026 (verified directly against the month).

C3. Which Weekdays Occur 5 Times

"In a 31-day month starting on a Tuesday, which weekdays occur 5 times?"
Method: §5 rule of Notes 02 — L = days − 28. For 31 days, $L = 3$: the starting weekday and the next two. Starting Tuesday $\Rightarrow$ **Tuesday, Wednesday, Thursday**.

C4. February-Specific Frequency

"In a leap year, which weekday occurs 5 times in February?"
Method: 29 days $\Rightarrow L = 1 \Rightarrow$ only the weekday 1 February falls on. In a non-leap February (28 days, $L = 0$), **no** weekday occurs 5 times — all seven occur exactly 4 times.

4 Category D — Same Calendar / Repeating Year

D1. Same Calendar as a Given Year

"The calendar for 2026 will be the same as the calendar for which future year?"
Method: Running-odd-day-total from Notes 02 §7 — don't assume a fixed gap. **Answer for 2026 specifically: 2037** (verified; 2026 is ordinary, gap = 11 years).

D2. Odd-One-Out Calendar

"Which of these years has a calendar different from the others: 1997, 2003, 2014, 2025?"
Method: Two years share a calendar only if both are the same type (ordinary/leap) **and** 1 January falls on the same weekday. Check leap/non-leap status first to eliminate options fast, then compare 1 January weekdays only among years of matching type.

D3. Calendar Repeats After N Years (Cycle Length)

"After how many years does a leap year's calendar typically repeat?"
Method: Table in Notes 02 §7 — typically **28 years** for a leap year (rarely 40, only near a century that breaks the 4-year leap rule). For an ordinary year, typically **6 or 11 years** (rarely 12). Never assume 17 — that number doesn't actually occur.

5 Category E — Leap Year Reasoning

E1. Is This Year a Leap Year?

"Is 2100 a leap year?"
Method: Notes 02 §4 quick check. 2100 is divisible by 4 and by 100, but **not** by 400 ⇒ **not a leap year**. Classic trap — most people stop at "divisible by 4."

E2. Counting Leap Years in a Range

"How many leap years are there from 1601 to 2000?"
Method: Notes 02 §4 formula, count(2000) − count(1600) = 485 − 388 = **97**.

E3. Century / Leap-Year Reasoning (CAT-style)

"Can the last day of a century (a year ending in 00) ever fall on a Tuesday?"
Method: Centuries only ever contribute 5, 3, 1, or 0 odd days (for 100, 200, 300, 400-year blocks) — never 2, 4, or 6. So a century's last day can only be **Sunday, Monday, Wednesday, or Friday**, never Tuesday, Thursday, or Saturday.

6 Category F — Date-to-Date and Span Problems

F1. Days Between Two Dates

"How many days are there between 15 August 2010 and 25 December 2010?"
Method: Convert both to a running day-of-year count (or subtract directly), being careful whether inclusive or exclusive is wanted. **Answer: 132 days** (exclusive of the start date).

F2. Consecutive Dates Within a Month

"If 18 January is a Tuesday, what day is 27 January?"
Method: Notes 02 §6 rule — difference in days mod 7. $27 - 18 = 9$, $9 \bmod 7 = 2 \rightarrow$ Tuesday $+2 =$ **Thursday**.

F3. Same Date, Different Year (Leap-Day Trap)

"If 15 March 2015 was a Sunday, what day was 15 March 2016?"
Method: Don't just add 1 odd day for "one year passing" — check whether 29 February falls **inside the specific span** being bridged. Here 29 Feb 2016 falls between the two dates, so the gap is **2** odd days, not 1. Sunday $+2 \rightarrow$ **Tuesday**.

7 Category G — Reverse Problems (Given a Clue, Find the Date/Day)

G1. Given the Nth Occurrence, Find the 1st Occurrence

"The third Saturday of a month falls on the 16th. Find the date of the first Saturday."
Method: Subtract $7 \times (N-1)$ days. Third minus 2 weeks: $16 - 14 = 2$. **Answer: 2nd.**

G2. Given the Month's Starting Day, Find a Later Date's Day

"A month starts on Wednesday and has 30 days. What day is the 30th?"
Method: Notes 02 §6 rule — $(30-1) \bmod 7 = 29 \bmod 7 = 1 \rightarrow$ Wednesday $+1 =$ **Thursday**.

G3. Calendar Matching — Find All Occurrences of a Weekday

"If the 15th of a month is a Friday, what other dates in that month are Fridays?"
Method: Every matching weekday is exactly 7 days apart. From the 15th: 1, 8, **15, 22, 29** (list depends on month length — always check both directions from the known date, not just forward).

8 Category H — Logic Puzzles and Word Problems

H1. Birthday / Recurrence Puzzles

"A person was born on a Monday. Is it possible for their birthday to fall on a Monday again every single year?"

Method: A birthday repeats on the same weekday only if the gap between birthdays contributes 0 odd days — impossible every year, since every ordinary year contributes 1 odd day and every leap year contributes 2. It can only repeat on the same weekday in the rare years where the running odd-day total (Notes 02 §7) resets to a multiple of 7, not annually.

H2. Working-Day Counting

"An office is closed every Sunday. If a 31-day month starts on a Saturday, how many working days does it have?"

Method: Count Sundays using the weekday-frequency rule (§C3/C4), subtract from total days. A 31-day month starting on Saturday has $L = 3$ extra weekdays (Sat, Sun, Mon occur 5 times) — so Sunday occurs **5** times ⇒ working days $= 31 - 5 = \mathbf{26}$.

H3. Multi-Clue Logic Puzzles

"1 January of a certain year is a Friday, and 1 January of the following year is a Sunday. What can you conclude about the certain year?"

Method: The shift from one 1 January to the next equals that year's odd-day contribution. Friday → Sunday is a shift of **2** odd days, which only a **leap year** contributes ⇒ the certain year must be a leap year.

9 Priority Order for Limited Prep Time

If you're short on time, drill these first — they cover the large majority of what actually appears in placement tests, roughly in order of frequency:

1. **A1** — Find the day for a given date (mod-400 or anchor method)
2. **B1/B2** — Day after/before N days (pure mod7)
3. **E1** — Leap year identification (the century exception is the #1 trap)
4. **C1** — Nth weekday of a month
5. **D1** — Same/repeating calendar year
6. **C3/C4** — Weekday frequency in 28/29/30/31-day months
7. **F1** — Days between two dates
8. **F3** — Same date, different year (leap-day-crossing trap)
9. **E3** — Century reasoning (CAT-style, less common but high-value when it appears)